Concept 5 | Applications of Integration to Geometry

English Student Edition • Focused formulas and guided practice

Formula opener

Area

$$A = \int_a^b (y_{top} - y_{bottom}) dx$$

Volume

$$V = \pi \int_a^b y^2 dx$$

Bounds

$$a \leq x \leq b$$

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Worked Solutions **Applications of Integration to Geometry**

Q001 Written

Find the area bounded by $y = 2x^2 + 1$, the x-axis, $x = -2$, and $x = 1$.

Solution

1. The parabola is above the x-axis on the interval.
2. $S = \int_{-2}^{1}(2x^2 + 1)\,dx = 9$.

Final answer

9

Q002 Written

Find the area bounded by $y = x^2 + x - 6$, the x-axis, $x = -2$, and $x = 1$.

Solution

1. The curve is below the x-axis on $[-2, 1]$.
2. Use $S = -\int_{-2}^{1}(x^2 + x - 6)\,dx = \frac{33}{2}$.

Final answer

$\frac{33}{2}$

Worked Solutions **Applications of Integration to Geometry**

Q003 Written

Find the area bounded by $y = -x^2 + 4$ and the x-axis.

Solution

1. The intercepts are $x = -2, 2$.
2. Integrate the positive curve from -2 to 2.

Final answer

$\frac{32}{3}$

Q004 Written

Find the area under $y = -x^2 - 2x + 3$ and above the x-axis.

Solution

1. The roots are -3 and 1.
2. Integrate $-x^2 - 2x + 3$ between the roots.

Final answer

$\frac{32}{3}$

Worked Solutions **Applications of Integration to Geometry**

Q005 **Written**

Find the area between $y = x^2 - 2$ and $y = -2x + 1$.

Solution

1. Intersections give $x = -3, 1$; the line is upper.
2. Integrate upper minus lower over $[-3, 1]$.

Final answer

$$\frac{32}{3}$$

Q006 **Written**

Find the area between $y = x^2 + 2x - 4$ and $y = -x^2 + 2x + 4$.

Solution

1. The intersections are $x = -2, 2$.
2. Use the upper-minus-lower integral $\int_{-2}^{2}(8 - 2x^2)\,dx$.

Final answer

$$\frac{64}{3}$$

Worked Solutions **Applications of Integration to Geometry**

Q007 Written

Find the area between $y = -x^2 + 4x$ and $y = x$.

Solution

1. Intersections are $x = 0, 3$.
2. Integrate $-x^2 + 3x$ from 0 to 3.

Final answer

$\frac{9}{2}$

Q008 Written

Find the area between $y = -x^2 + 8$ and $y = 2x$ for $-2 \le x \le 3$.

Solution

1. The curves meet at $x = 2$ within the interval.
2. Split the absolute difference at $x = 2$; the two areas are 80/3 and 10/3.

Final answer

30

Worked Solutions **Applications of Integration to Geometry**

Q009 Written

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x-axis.

Solution

1. Roots are (0,2,3); the curve changes sign at 2.
2. Add the positive area on $[0, 2]$ to the absolute area on $[2, 3]$.

Final answer

$\frac{37}{12}$

Q010 Written

Find the area bounded by $y = -x^3 + 4x$ and the x-axis.

Solution

1. Roots are (-2,0,2); the two lobes have equal area.
2. Double $\int_0^{2}(-x^3 + 4x)\,dx = 4$.

Final answer

8

Worked Solutions

Applications of Integration to Geometry

Q011 **Written**

Solve the given cubic area exercises by finding every x-intercept and adding the absolute signed pieces.

Solution

1. Factor the cubic first.
2. Mark the sign on each interval before integrating.

Final answer

Use the numbered solution for each given cubic.

Q012 **Written**

Evaluate $\int_{-2}^{1} |x+1|\, dx$.

Solution

1. The sign changes at $x = -1$.
2. Split into two ordinary integrals and add the positive areas.

Final answer

$\frac{5}{2}$

Worked Solutions **Applications of Integration to Geometry**

Q013 **Written**

Evaluate $\int_0^3 |x^2 - 4|\, dx$.

Solution

1. The root in the interval is $x = 2$.
2. Use $-\int_0^{\,2}(x^2 - 4)\, dx + \int_2^{\,3}(x^2 - 4)\, dx$.

Final answer

$\frac{23}{3}$

Q014 **Written**

Evaluate $\int_0^2 |x^2 + 3x - 4|\, dx$.

Solution

1. The relevant root is $x = 1$.
2. Reverse the sign on $[0, 1]$, then add the integral on $[1, 2]$.

Final answer

5

Worked Solutions **Applications of Integration to Geometry**

Q015 Written

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.

Solution

1. The tangents are $y = -x$ and $y = 3x - 4$, meeting at $x = 1$.
2. Split the area at $x = 1$ and integrate curve minus tangent.

Final answer

$\frac{3}{2}$

Q016 Written

Find the area bounded by $y = x^3 + x^2 - 3x + 6$ and its tangent at $(1, 5)$.

Solution

1. The tangent is $y = 2x + 3$; the intersections are $x = -3, 1$.
2. Factor the difference as $(x - 1)^2(x + 3)$ and integrate.

Final answer

$\frac{64}{3}$

Worked Solutions **Applications of Integration to Geometry**

Q017 Written

For $y = x^2$, tangents through $(3, 8)$ touch at $x = 2, 4$. Find the enclosed area.

Solution

1. The tangent equations are $y = 4x - 4$ and $y = 8x - 16$.
2. Integrate the two squared gaps on $[2, 3]$ and $[3, 4]$.

Final answer

$$\frac{2}{3}$$

Q018 Written

If $y = kx$ bisects the area under $y = 2x - x^2$ and above the x-axis, find k.

Solution

1. The total area under the parabola is 3.
2. Set the area cut off by $y = kx$ equal to $3/2$, giving $(2 - k)^3 = 3/4$.

Final answer

$$k = 2 - \sqrt[3]{\frac{3}{4}}$$

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Q019 Written

Solve the given bisection exercises for the given parabolas and lines, using the required area ratio.

Solution

1. Find the total parabola area first.
2. Integrate between the line and curve, then impose the stated ratio.

Final answer

Use the numbered solution for each given ratio.

Q020 MCQ

When the graph is below the x-axis, area is:

Solution

1. Reverse the sign.

Correct option: B

Final answer

negative of the integral

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Q021 MCQ

A bounded area under a curve is found by first checking:

Solution

1. Draw the graph.

Correct option: A

Final answer

the graph and sign

Q022 MCQ

The area under $y = -x^2 + 4$ between its roots is:

Solution

1. Integrate from -2 to 2.

Correct option: C

Final answer

32/3

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Q023 MCQ

Between two curves, the integrand is:

Solution

1. Identify the upper graph.

Correct option: B

Final answer

upper minus lower

Q024 MCQ

If curves cross inside the interval, area must be:

Solution

1. The sign of the difference changes.

Correct option: A

Final answer

split at the crossing

Worked Solutions **Applications of Integration to Geometry**

Q025 MCQ

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

Solution

1. Use the intersection limits.

Correct option: B

Final answer

32/3

Q026 MCQ

For cubic area, the sign changes at:

Solution

1. Mark all roots.

Correct option: B

Final answer

each x-intercept

Worked Solutions **Applications of Integration to Geometry**

Q027 MCQ

The area for $y = -x^3 + 4x$ is:

Solution

1. The two lobes are equal.

Correct option: C

Final answer

8

Q028 MCQ

A negative lobe contributes:

Solution

1. Area is non-negative.

Correct option: B

Final answer

positive absolute area

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Q029 **MCQ**

For an absolute-value integral, first locate:

Solution

1. Roots set the sign intervals.

Correct option: A

Final answer

the roots

Q030 **MCQ**

The value of $\int_0^2 |x^2+3x-4| dx$ is:

Solution

1. Split at $x = 1$.

Correct option: C

Final answer

5

Worked Solutions **Applications of Integration to Geometry**

Q031 MCQ

If f changes sign at c, split the integral:

Solution

1. The absolute value changes form.

Correct option: A

Final answer

atc

Q032 MCQ

A tangent line at x=a has slope:

Solution

1. Differentiate the curve.

Correct option: B

Final answer

$f'a$

Worked Solutions **Applications of Integration to Geometry**

Q033 MCQ

For two tangent lines, split at their:

Solution

1. The upper tangent changes there.

Correct option: A

Final answer

intersection x-coordinate

Q034 MCQ

The area in the given parabola-tangent Warm Up is:

Solution

1. Integrate the two gaps.

Correct option: C

Final answer

3/2

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Q035 MCQ

Bisection means each part has:

Solution

1. Set each part to half the total.

Correct option: A

Final answer

the same area

Q036 MCQ

For $y=2x-x^2$, the total area above the x-axis is:

Solution

1. Integrate from 0 to 2.

Correct option: C

Final answer

3

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Q037 MCQ

The line y=kx must satisfy the stated:

Solution

1. Use the given bisection condition.

Correct option: A

Final answer

area ratio

Q038 MCQ

In a bisection problem, the cut-off area is:

Solution

1. Set the two parts equal.

Correct option: B

Final answer

half the total area

Worked Solutions **Applications of Integration to Geometry**

Quiz MCQ 1 **Concept Quiz · MCQ**

The area under $y = -x^2 + 4$ and above the x-axis is:

Solution

1. Integrate from -2 to 2.

Correct option: C

Final answer

$$\frac{32}{3}$$

Quiz MCQ 2 **Concept Quiz · MCQ**

The area between $y = x^2 - 2$ and $y = -2x + 1$ is:

Solution

1. Use upper minus lower from -3 to 1.

Correct option: B

Final answer

$$\frac{32}{3}$$

Worked Solutions **Applications of Integration to Geometry**

Quiz MCQ 3 Concept Quiz · MCQ

Evaluate $\int_{-2}^{1} |x+1|\,dx$:

Solution

1. Split at $x = -1$.

Correct option: B

Final answer

$\frac{5}{2}$

Quiz MCQ 4 Concept Quiz · MCQ

The area enclosed by $y = x^2$ and tangents through $(3, 8)$ is:

Solution

1. Integrate the two squared gaps.

Correct option: C

Final answer

$\frac{2}{3}$

Worked Solutions **Applications of Integration to Geometry**

Quiz WRITTEN 5 **Concept Quiz · Written**

Find the area bounded by $y = x^3 - 5x^2 + 6x$ and the x-axis.

Solution

1. Use roots (0,2,3) and split at 2.

Final answer

$\frac{37}{12}$

Quiz WRITTEN 6 **Concept Quiz · Written**

Evaluate $\int_0^2 |x^2 + 3x - 4|\, dx$.

Solution

1. Split at the root $x = 1$.

Final answer

5

Worked Solutions **Applications of Integration to Geometry**

Quiz WRITTEN 7 Concept Quiz · Written

Find the area bounded by $y = x^2 - x$ and its tangents at $(0, 0)$ and $(2, 2)$.

Solution

1. Find both tangents and split at their intersection.

Final answer

$$\frac{3}{2}$$

Quiz WRITTEN 8 Concept Quiz · Written

If $y = kx$ bisects the area under $y = 2x - x^2$, find k.

Solution

1. Set the cut-off area equal to half of 3.

Final answer

$$2 - \sqrt[3]{\frac{3}{4}}$$

Answer key

Use the question number shown on each page.

Q001 9

Q002 $\frac{33}{2}$

Q003 $\frac{32}{3}$

Q004 $\frac{32}{3}$

Q005 $\frac{32}{3}$

Q006 $\frac{64}{3}$

Q007 $\frac{9}{2}$

Q008 30

Q009 $\frac{37}{12}$

Q010 8

Q011 Use the numbered solution for each given cubic.

Q012 $\frac{5}{2}$

Q013 $\frac{23}{3}$

Q014 5

Q015 $\frac{3}{2}$

Q016 $\frac{64}{3}$

Q017 $\frac{2}{3}$

Q018 $k = 2 - \sqrt[3]{\frac{3}{4}}$

Q019 Use the numbered solution for each given ratio.

Q020 B

Q021 A

Q022 C

Q023 B

Q024 A

Q025 B

Q026 B

Q027 C

Q028 B

Q029 A

Q030 C

Q031 A

Q032 B

Q033 A

Q034 C

Q035 A

Q036 C

Q037 A

Q038 B

Quiz MCQ 1 C

Quiz MCQ 2 B

Quiz MCQ 3 B

Quiz MCQ 4 C

Quiz WRITTEN 5 $\frac{37}{12}$

Quiz WRITTEN 6 5

Quiz WRITTEN 7 $\frac{3}{2}$

Quiz WRITTEN 8 $2 - \sqrt[3]{\frac{3}{4}}$